

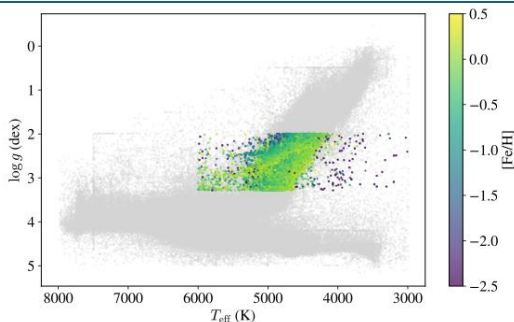
Asteroseismology of Red Giants: Discovering the Origins of Li-rich Giants

Alexa C. Beyer¹, Jamie Tayar¹, Rafael Garcia², Leslie Morales¹, Danil?, Lina?



Why are Li-rich Giants Weird ?

Lithium, as an easily destroyed element, provides a sensitive tracer for the interiors of low-mass stars. From stellar evolution models, we expect from the PMS and MS destruction of lithium and dilution from post-MS first dredge-up, we expect lithium abundances for giants to be $A(\text{Li}) < 1.5$ dex. Observations have found $< 1\%$ of giants are given the title of "Li-rich".

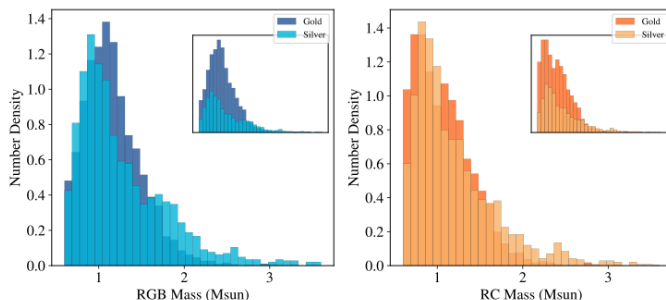


Data:

We collect a sample of 23,472 red giants observed by TESS from the Wang et al. 2023 3D NLTE sample of lithium abundances. Our pySYD returned asteroseismology for 10,597 stars with $T_{\text{mag}} < 12.8$ and 5 or more sectors of data.

Seismology:

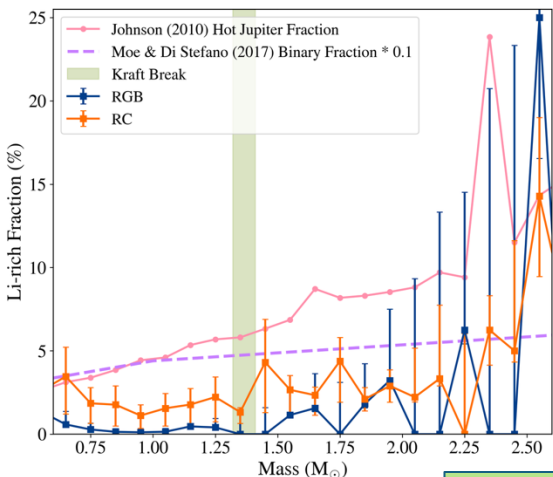
We use pySYD and inputting TESS light curves to apply a Fourier transform to detect the region of excess frequency to determine the maximum power (ν_{max}) and the large frequency spacing ($\Delta\nu$).



Masses and Ages:

To produce our final mass and radius values, we use the scaling relations with correction factors for 3,583 stars in our gold sample using ν_{max} and $\Delta\nu$, 1,763 silver using ν_{max} and *Gaia* radii, and 4,558 bronze using seismology informed models. We also compute uncertainties by propagating individual uncertainties on ν_{max} , $\Delta\nu$, effective temperature, and $\log g$ through seismic scaling relations using traditional error propagation.

How do giants become lithium rich ?

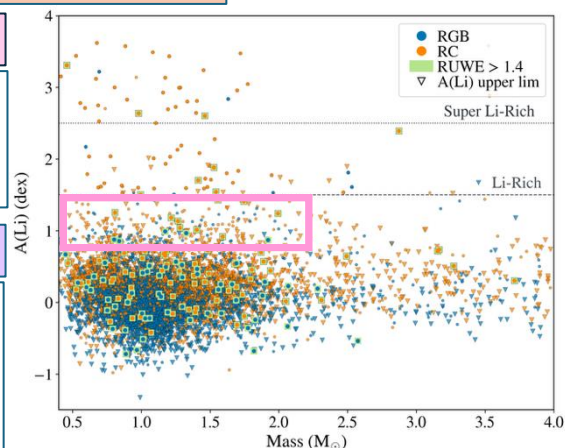


Planetary Engulfment

Some amount of a planet's abundance dissolves through the surface of an expanding red giant. We investigate the fraction of hot/warm planets as a function of mass.

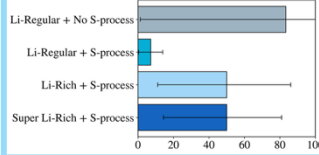
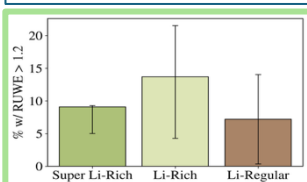
Binary Merger

The companion fraction among lower mass stars shows us a similar trend increase rates at higher mass, but this does not discretely account for all Li-Rich stars are the result of binaries, otherwise we would see a higher % Li-rich stars.



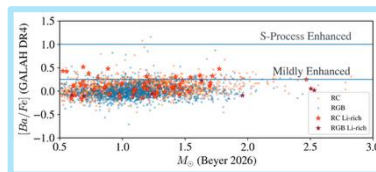
Binary Interaction

Binary fraction accounts for a small fraction of our li-rich sample, comparable to the trend of binary fraction as a function of mass.



AGB Companion

If theorizing an AGB star companion, along with Li enrichment, we expect to find in above average abundance of S-process elements (ex: Ba).



CONCLUSIONS:

- The binary fraction can account for some but not all li-rich giants via merger tidal interactions. Rotation is harder to measure and incomplete.
- S-process enrichment indicates possible AGB star accretion onto a companion.
- Not enough RGB Li-Rich stars to attribute to planetary engulfment.